



ROADGUARD AI

The domain of the Project:
Generative AI

Team Mentors:

Mr. Prujith Radhakrishnan
Project Mentor SURE
Trust

Team Members:

Ms. Kinjel Sharma

Period of the project

January 2026 to July 2026



Declaration

The project titled “**RoadGuard AI**” has been mentored by **Mr. Prujith Radhakrishnan**, organised by **SURE Trust**, from January 2026 to July 2026, for the benefit of educated unemployed youth for gaining hands-on experience in working on industry relevant projects that would take them closer to prospective employment.

I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Member

Ms. Kinjel Sharma

Signature

Mentor

Mr. Prujith Radhakrishnan
Project Mentor
SURE Trust

Prof. Radhakumari
Executive Director & Founder
SURE Trust



Table of Contents

1. Executive Summary
2. Introduction
3. Project Objectives
4. Methodology and Results
5. Learning and Reflection
6. Conclusion and Future Scope



Executive Summary

RoadGuard AI is an intelligent vehicle breakdown assistance and insurance claim support platform developed to help vehicle owners make informed decisions during roadside emergencies and after road accidents. The project combines Computer Vision, Generative Artificial Intelligence, Large Language Models (LLMs), and Multi-Agent AI systems to analyse vehicle images, understand accident scenarios, interpret insurance policy documents, and provide personalised recommendations.

The primary objective of RoadGuard AI is to minimise confusion and delay during emergency situations by providing instant AI-powered assistance. The system first analyses uploaded vehicle images and determines whether the problem can be repaired immediately, whether the vehicle can safely reach a nearby garage, or whether towing assistance is required. For accident cases, the application evaluates the extent of vehicle damage, analyses the insurance policy, estimates the claim strength, predicts settlement probability, and provides a structured claim strategy for the vehicle owner.

The project has been developed using a modern full-stack architecture consisting of React.js for the frontend, FastAPI for backend services, MongoDB as the database, and Google's Gemini Vision model together with Groq Large Language Models for multimodal reasoning. Retrieval-Augmented Generation (RAG) techniques and multiple AI agents are incorporated to improve decision making and produce reliable recommendations.

The Vehicle Breakdown Assistance module categorises roadside situations into three intelligent decision cases:

- Roadside repair using available tools.
- Temporary repair followed by navigation to the nearest authorised garage.
- Immediate towing assistance when driving is unsafe.

The Insurance Intelligence module acts as an AI-powered claim advisor. It analyses accident photographs, user descriptions, and insurance policy documents before generating an executive summary, coverage analysis, claim strategy, negotiation strategy, documentation checklist, and a comprehensive downloadable PDF report. A multi-agent workflow consisting of Claim Maximizer, Insurance Reviewer, and Judge agents ensures that recommendations remain practical, balanced, and legally reasonable.



RoadGuard AI demonstrates the practical application of Generative AI in solving real-world transportation and insurance challenges. The system reduces response time during emergencies, assists users in understanding insurance policies, improves transparency in claim settlement, and provides actionable recommendations that help maximise legitimate insurance claims. The project highlights the integration of Artificial Intelligence, Computer Vision, Natural Language Processing, and full-stack web technologies into a single intelligent platform capable of assisting vehicle owners throughout the complete roadside assistance and insurance claim lifecycle.



Introduction

1. Background and Context of the Project

Roadside vehicle breakdowns and insurance claim procedures are often stressful, time-consuming, and confusing for vehicle owners. Drivers frequently struggle to identify the actual fault in their vehicle, determine whether it is safe to continue driving, locate nearby repair facilities, or understand the insurance claim process after an accident. Recent advancements in Artificial Intelligence, Computer Vision, and Large Language Models provide an opportunity to automate these tasks and offer intelligent assistance in real time. RoadGuard AI was developed to address these practical challenges by combining image analysis with Generative AI.

2. Problem Statement

Most existing roadside assistance applications only provide emergency contact numbers or towing services without analysing the actual condition of the vehicle. Similarly, insurance claim processes require users to manually interpret policy documents and prepare supporting evidence. This project aims to build an AI-powered platform capable of analysing vehicle damage, recommending the correct roadside action, and providing intelligent insurance claim guidance using uploaded accident images and policy documents.

3. Scope and Limitations

The scope of RoadGuard AI includes vehicle breakdown diagnosis, roadside assistance recommendations, garage and towing suggestions, insurance policy analysis, accident image assessment, claim estimation, negotiation strategy generation, and downloadable report generation. The system currently focuses on passenger vehicles and relies on clear images and accurate user-provided information. The recommendations generated are advisory in nature and should not replace professional mechanical inspection or legal consultation.

4. Innovation in the Project

The key innovation of RoadGuard AI is the integration of Computer Vision with a Multi-Agent Generative AI architecture. Instead of providing generic advice, the system analyses uploaded vehicle images, classifies the severity of the damage, predicts the most appropriate roadside action, evaluates insurance coverage, estimates claim settlement values, and generates negotiation strategies through specialised AI agents including the Claim Maximizer Agent, Insurance Reviewer Agent, and Judge Agent. This combination



Innovation&EntrepreneurshipHubforEducatedRuralYouth(SURETrust-IERY)

enables intelligent decision support for both roadside emergencies and insurance claim management within a single web platform.

Project Objectives

The primary objective of RoadGuard AI is to develop an intelligent AI-powered platform capable of assisting vehicle owners during roadside breakdowns and insurance claim procedures. The system combines Artificial Intelligence, Computer Vision, and Generative AI technologies to provide accurate recommendations, reduce emergency response time, and simplify insurance claim processing.

The specific objectives of this project are listed below:

1. Develop an AI-based roadside assistance system capable of analysing uploaded vehicle images and identifying the severity of the breakdown.
2. Classify vehicle breakdowns into three intelligent decision categories:
 - Roadside Repair
 - Temporary Repair with Nearby Garage Recommendation
 - Towing Assistance
3. Provide users with clear repair guidance, nearby garage suggestions, and towing recommendations based on the detected vehicle condition.
4. Build an AI-powered Insurance Claim Intelligence module that analyses accident photographs, user descriptions, and insurance policy documents.
5. Automatically evaluate insurance coverage, identify covered damages, estimate claim settlement values, and generate claim strength assessments.
6. Implement a Multi-Agent AI architecture consisting of specialised reasoning agents such as the Claim Maximizer Agent, Insurance Reviewer Agent, and Judge Agent to improve the reliability and quality of claim recommendations.
7. Generate a comprehensive downloadable PDF report containing claim analysis, negotiation strategies, documentation checklist, settlement estimation, and recommended next actions.
8. Design a modern and user-friendly web interface that enables users to upload vehicle images, insurance documents, and accident descriptions with minimal effort.



9. Demonstrate the practical application of Generative AI, Computer Vision, RetrievalAugmented Generation (RAG), and Large Language Models in solving real-world transportation and insurance problems.
10. Develop a scalable and modular architecture that can be extended in the future with additional AI models, regional insurance policies, multilingual support, and predictive vehicle maintenance capabilities.

Methodology & Results

4.1 Methods / Technology Used

RoadGuard AI was developed as a full-stack Artificial Intelligence based web application for assisting vehicle owners during roadside emergencies and insurance claim processing. The project combines modern web technologies, cloud-based AI models, computer vision, and intelligent decision-making techniques.

The application consists of two major modules:

- Vehicle Breakdown Assistance
- Insurance Claim Intelligence

The frontend provides an interactive dashboard through which users upload vehicle images and accident details. The backend processes the uploaded information, communicates with AI models, performs analysis, and generates intelligent recommendations.

The system follows a modular architecture where each component performs a specific responsibility including authentication, image processing, AI reasoning, report generation, and database management.



4.2 Tools / Software Used

The following technologies were used during the development of RoadGuard AI.

Technology	Purpose
React.js	Frontend User Interface
FastAPI	Backend REST API
MongoDB	Database Storage
Google Gemini Vision	Vehicle Image Analysis
Groq LLM	Claim Reasoning and Report Generation
Python	Backend Development
JavaScript	Frontend Development
HTML/CSS	User Interface Design
Git	Version Control
Visual Studio Code	Development Environment
Postman	API Testing

The combination of these technologies enables RoadGuard AI to analyze images, understand insurance policies, provide intelligent roadside recommendations, and generate professional claim reports.



4.3 Data Collection Approach

The project does not rely on a fixed training dataset. Instead, it processes real-time user inputs for every request.

The following information is collected during execution:

- Vehicle photographs uploaded by the user.
- Accident photographs showing damaged components.
- Insurance policy document (PDF).
- User description explaining the incident.
- Vehicle information.
- Location information for garage recommendation.

The uploaded images are analysed using Google Gemini Vision, while textual reasoning and claim optimization are performed using Large Language Models. This approach allows the system to generate dynamic recommendations for every individual case instead of relying on predefined rules.



4.4 Project Architecture

RoadGuard AI follows a modular client-server architecture.

The frontend is developed using React and communicates with the FastAPI backend through REST APIs.

The backend performs authentication, validates uploaded files, processes images using Gemini Vision, performs reasoning using Groq LLM, stores records in MongoDB, and generates PDF reports.

The architecture consists of the following components:

- User Interface
- Authentication Module
- Vehicle Breakdown Module
- Insurance Claim Module
- AI Processing Layer
- Database Layer
- PDF Report Generator

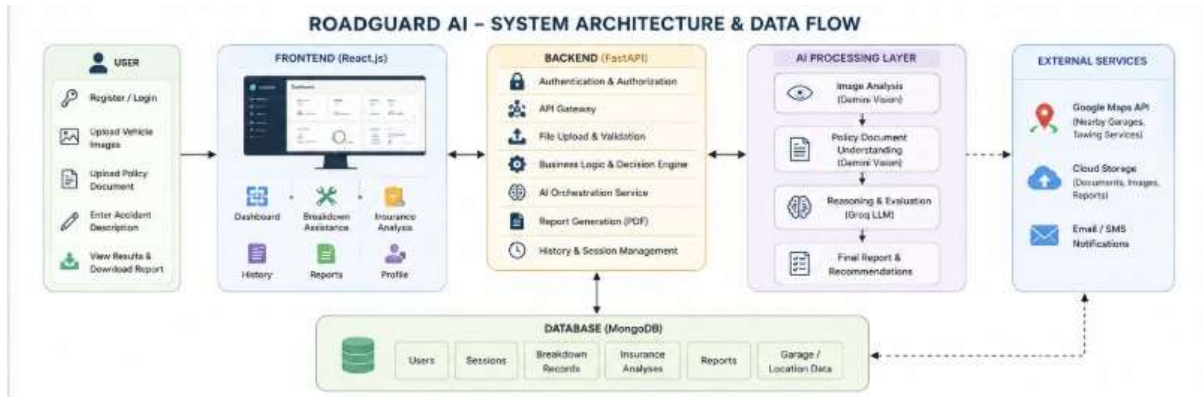


Figure 1: Overall Project Architecture

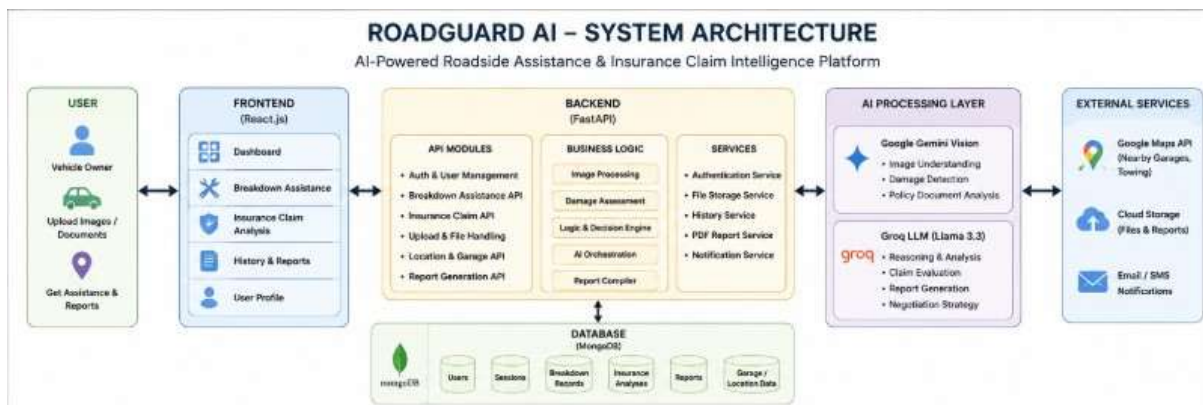


Figure 2: System Architecture of RoadGuard AI



4.5 Project Working Screenshots

The following screenshots demonstrate the working of different modules implemented in RoadGuard AI.

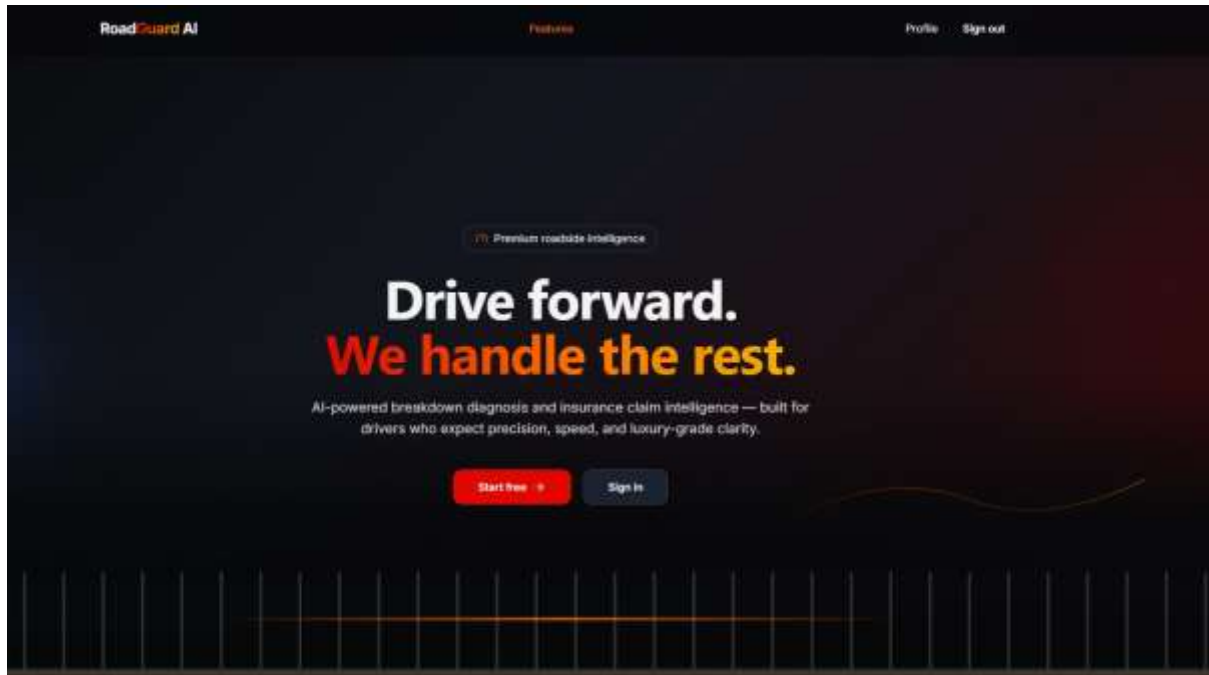


Figure 3: RoadGuard AI Dashboard

The dashboard provides quick navigation to all modules including Breakdown Assistance, Insurance Analysis, History, Reports, and User Profile.

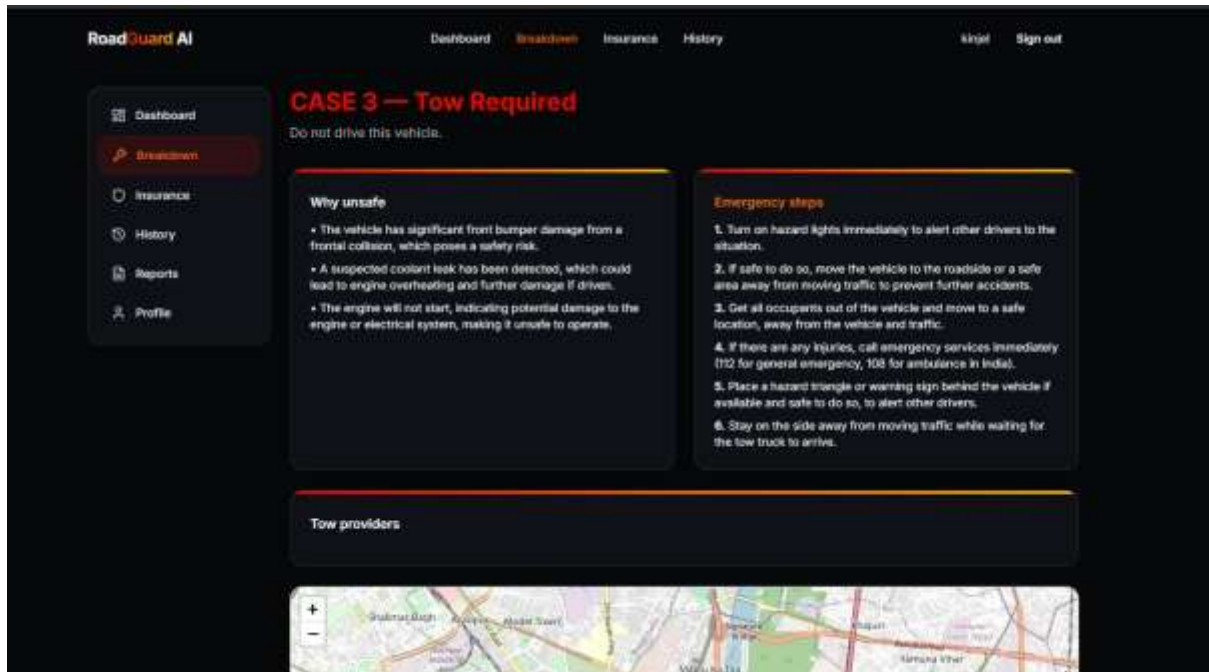


Figure 4: Vehicle Breakdown Assistance - Towing Required

This module analyses uploaded vehicle photographs and classifies the situation into roadside repair, temporary repair with nearby garage recommendation, or towing assistance.

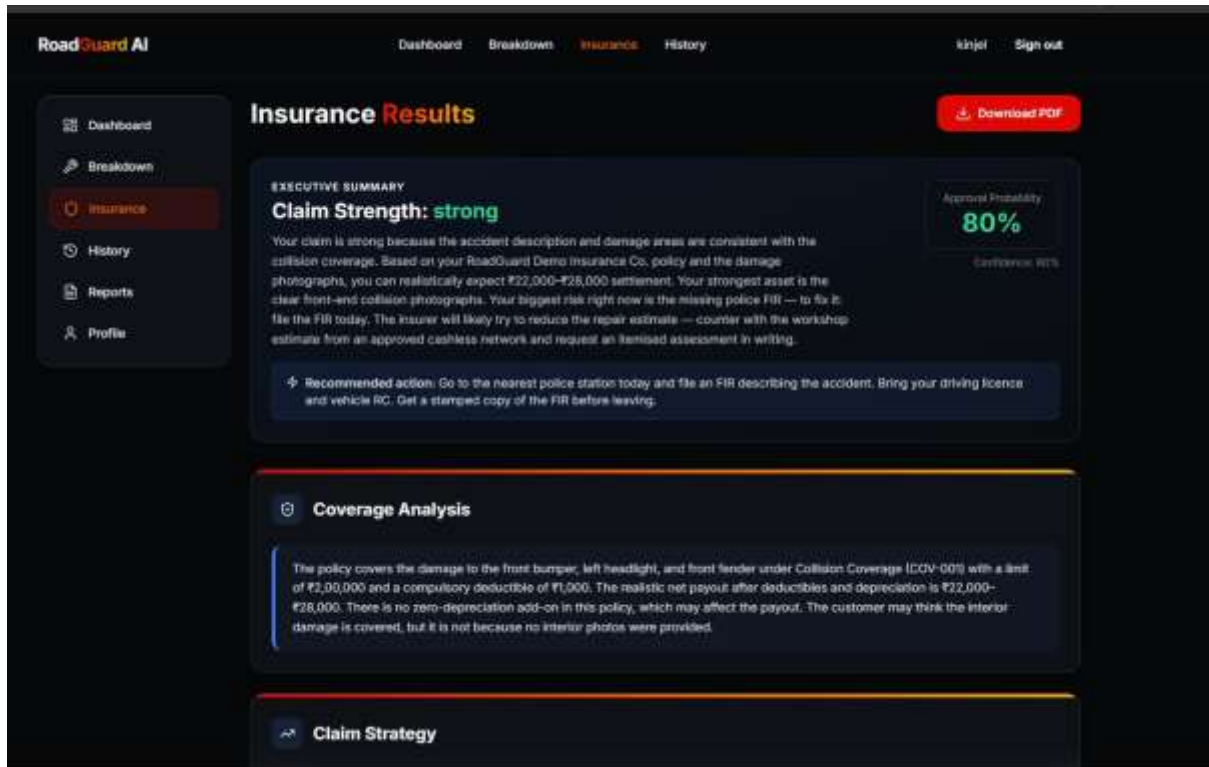


Figure 5: Insurance Executive Summary

The executive summary provides the overall claim strength, estimated approval probability, expected settlement range, and recommended immediate action.

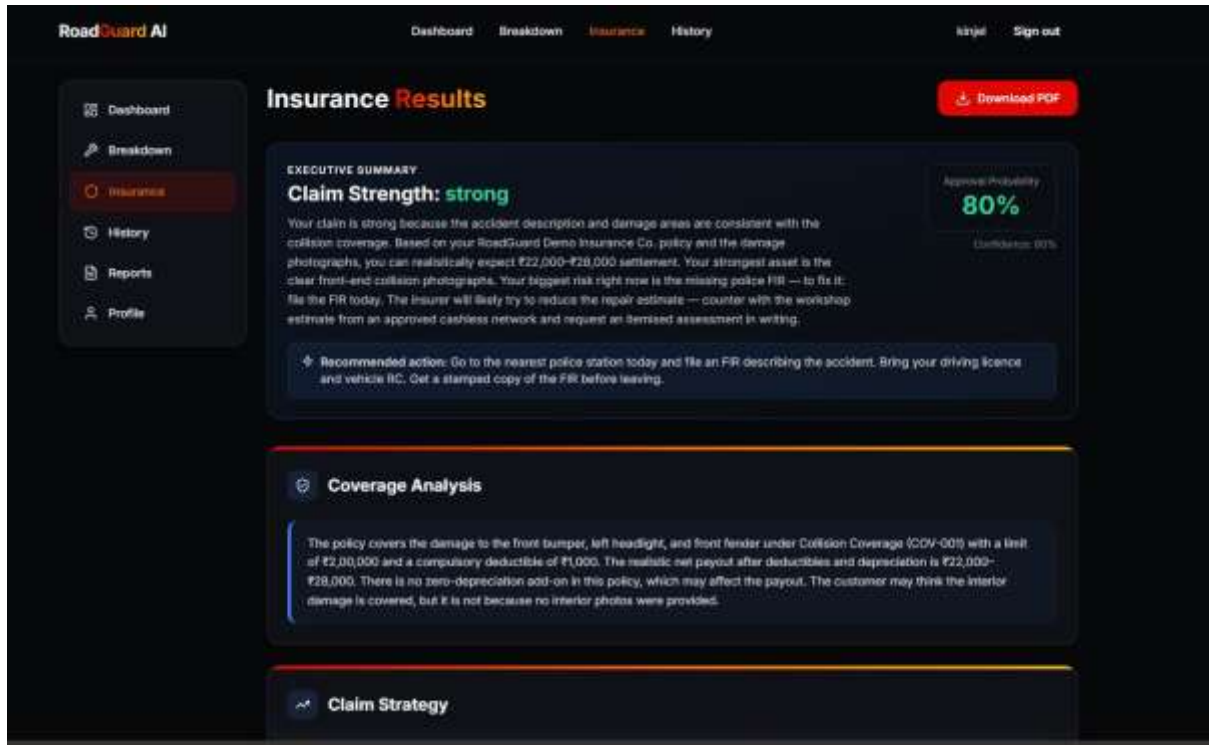


Figure 6: Coverage Analysis

The system analyses the insurance policy, identifies covered damages, estimates repair costs, and recommends an optimized claim strategy.

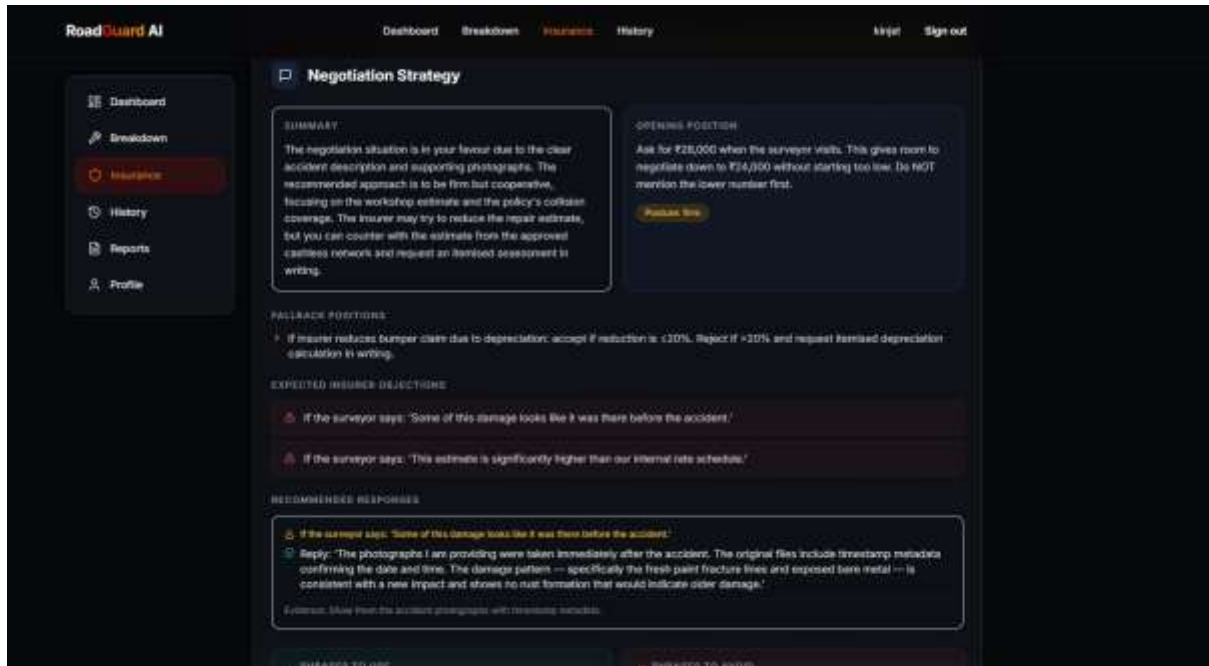


Figure 7: Negotiation Strategy

The AI recommends negotiation techniques, expected insurer objections, and suitable responses to improve the possibility of claim approval.

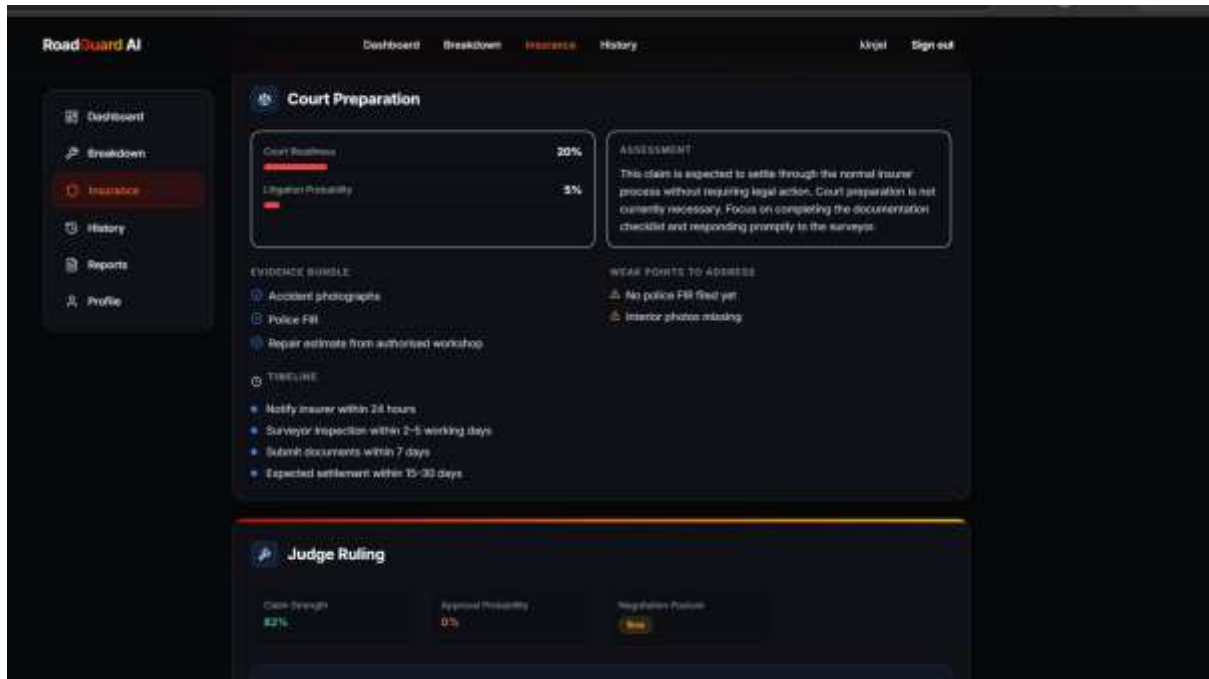


Figure 8: Court Preparation

The final AI analysis evaluates evidence strength, litigation readiness, missing documents, and provides the final judgement regarding claim validity.

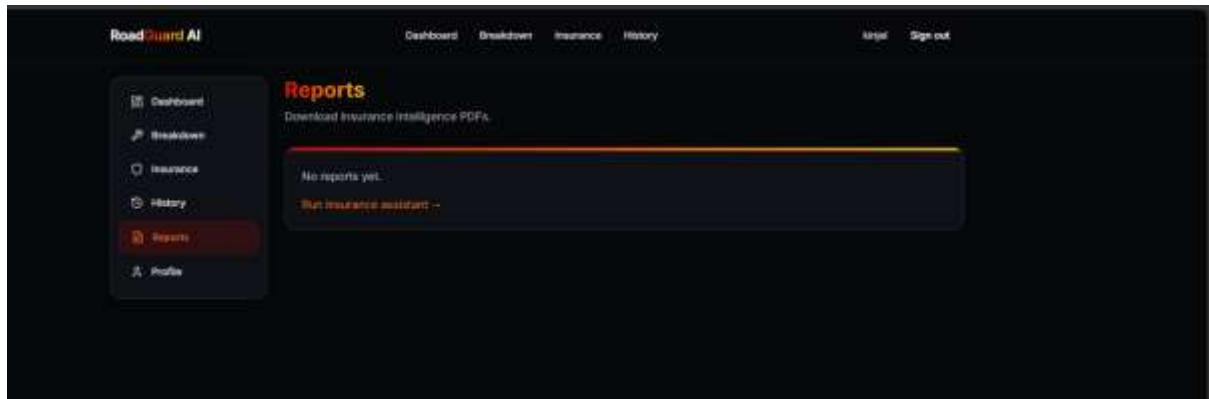


Figure 9: Reports Module

The History module stores previous analyses, the Reports module enables downloading generated PDF reports, and the Profile module manages user account information.



4.6 Project GitHub Repository

The complete source code of RoadGuard AI is maintained using Git version control. **GitHub Repository:**

<https://github.com/sure-trust/KINJEL-SHARMA-g4-gen-ai/tree/main/Final%20capstone%20project/RoadGuard-AI>

The repository contains the frontend, backend, AI integration modules, database configuration, API documentation, and project setup instructions.

The use of GitHub enables efficient version management, collaborative development, and secure source code backup throughout the project lifecycle.

Learning and Reflection

5.1 Technical Learning

During the development of RoadGuard AI, I gained practical experience in designing and implementing a full-stack Artificial Intelligence application. I strengthened my understanding of modern web development using React, FastAPI, and MongoDB while integrating cloud-based AI models such as Google Gemini Vision and Groq LLM.

I learned how to process real-world user inputs, including vehicle images, insurance policy documents, and textual accident descriptions. This project enhanced my knowledge of REST APIs, backend development, image processing workflows, prompt engineering, and AI-based decision-making systems. I also gained experience in generating dynamic PDF reports, maintaining session history, handling authentication, and building scalable software architecture.

5.2 Overall Experience

This internship provided valuable exposure to solving practical industry problems using Artificial Intelligence. Working on RoadGuard AI improved my problem-solving ability, software design skills, and confidence in developing end-to-end AI applications.

Throughout the project, I learned the importance of systematic planning, modular software architecture, debugging, testing, and continuous improvement. Integrating multiple technologies into a single application helped me understand how real-world software products are developed and deployed.

The internship also improved my technical documentation, project management, and independent learning skills. Overall, the experience significantly enhanced my practical knowledge of Generative AI, Computer Vision, full-stack development, and intelligent automation, preparing me for future software engineering and AI development roles.



Conclusion and Future Scope

6.1 Conclusion

RoadGuard AI was successfully designed and developed as an intelligent web-based platform that combines Artificial Intelligence, Computer Vision, and modern web technologies to assist vehicle owners during roadside emergencies and insurance claim processing.

The project successfully achieved its primary objectives of developing a smart vehicle breakdown assistance system and an AI-powered insurance claim assistant. The system can analyse uploaded vehicle images, classify roadside situations, recommend suitable actions such as roadside repair, garage assistance, or towing, and generate detailed insurance claim reports based on accident photographs, policy documents, and user descriptions.

During the development process, various technologies including React, FastAPI, MongoDB, Google Gemini Vision, and Groq LLM were integrated into a single application. The project also provides user authentication, session history, downloadable PDF reports, and an interactive dashboard, making it a complete end-to-end solution.

Overall, RoadGuard AI demonstrates how Generative AI can simplify complex decisionmaking processes, improve user experience, and provide practical assistance in real-world automotive and insurance scenarios.

6.2 Project Achievements

- Successfully developed a complete full-stack AI application.
- Implemented Vehicle Breakdown Assistance using image analysis.
- Developed an AI-powered Insurance Claim Intelligence module.
- Integrated Google Gemini Vision for vehicle damage assessment.
- Implemented intelligent claim analysis and report generation using Large Language Models.
- Designed a responsive dashboard with authentication, history, reports, and profile management.

- Generated downloadable PDF reports containing claim strength, settlement estimation, negotiation strategy, and documentation checklist.
- Improved practical understanding of Generative AI, Computer Vision, REST APIs, and full-stack software development.

6.3 Future Scope

Although RoadGuard AI successfully achieves its current objectives, several enhancements can further improve the system in future versions.

- Integrate real-time GPS tracking to automatically locate nearby service centres, garages, hospitals, and towing providers.
- Support live video-based vehicle inspection for more accurate damage assessment.
- Integrate with official insurance company APIs for automatic claim submission and status tracking.
- Add multilingual voice assistance to improve accessibility for users across different regions.
- Develop Android and iOS mobile applications for easier roadside access.
- Implement predictive vehicle maintenance using historical service records and IoT sensor data.
- Improve AI accuracy by training custom vehicle damage detection models on larger datasets.
- Add emergency SOS features for automatically notifying family members and emergency services during severe accidents.
- Expand support for commercial vehicles, electric vehicles, and two-wheelers.
- Deploy the system on cloud infrastructure for large-scale real-time usage.

In conclusion, RoadGuard AI demonstrates the practical application of Generative AI in solving real-world transportation and insurance challenges. The project provides an intelligent, user-friendly, and scalable solution that has significant potential for further research, industrial deployment, and future enhancement.